HUST

ĐẠI HỌC BÁCH KHOA HÀ NỘI
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.

Discrete Mathematics

PART 1

COMBINATORIAL THEORY

(Lý thuyết tổ hợp)

PART 2

GRAPH THEORY

(Lý thuyết đồ thị)

Content of Part 2

Chapter 1. Fundamental concepts

Chapter 2. Graph representation

Chapter 3. Graph Traversal

Chapter 4. Tree and Spanning tree

Chapter 5. Shortest path problem

Chapter 6. Maximum flow problem

Graph Representation

1. Incidence matrix
2. Adjacency matrix
3. Weight matrix
4. Adjacency list

Graph Representation

1. **Incidence matrix**
2. Adjacency matrix
3. Weight matrix
4. Adjacency list

1. Incidence Matrix

G = (V, E) is an unditected graph:

- V = {v_1, v_2, v_3, ..., v_n }
- E = {e_1, e_2, ..., e_m }

Then the incidence matrix with respect to this ordering of V and E is the *n* x *m* matrix M = [m_{ij}], where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i \\ 0 & \text{otherwise} \end{cases}$$

Can also be used to represent :

- **Multiple edges:** by using columns with identical entries, since these edges are incident with the same pair of vertices
- **Loops:** by using a column with exactly one entry equal to 1, corresponding to the vertex that is incident with the loop

1.Incidence Matrix

Matrix $M_{|V| \times |E|} = [m_{ij}]$, where

$$m_{ij} = \begin{cases} 1 & \text{when edge } e_j \text{ is incident with } v_i \\ 0 & \text{otherwise} \end{cases}$$

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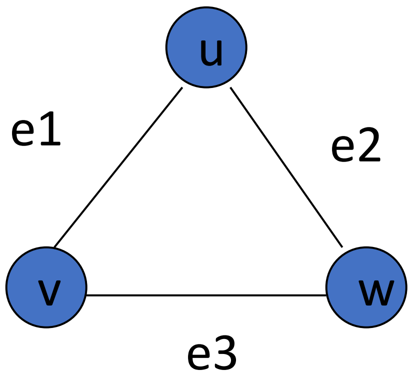
1.Incidence Matrix

Example: G = (V, E)



	e_1	e_2	e_3
v	1	0	1
u	1	1	0
w	0	1	1

Graph Representation

1. Incidence matrix
2. **Adjacency matrix**
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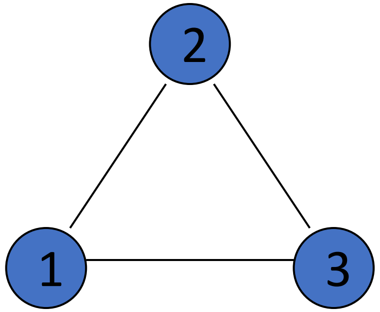
2. Adjacency Matrix

The Adjacency Matrix (NxN) $A = [a_{ij}]$ where $|V| = N$

For undirected graph

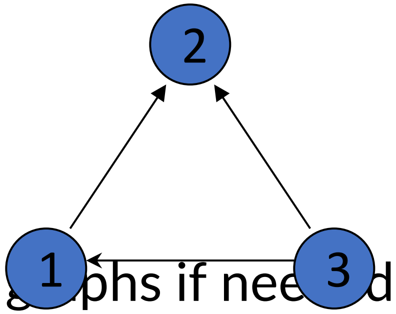
$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$



$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

For directed graph

$$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \text{ is an edge of } G \\ 0 & \text{otherwise} \end{cases}$$



$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}$$

This makes it easier to find subgraphs, and to reverse graphs if needed.

Graph Representation

1. Incidence matrix
2. Adjacency matrix
3. **Weight matrix**
4. Adjacency list

3. Weight matrix

- **Weighted** graphs have values associated with edges.
- In the case weighted graphs, instead of adjacency matrix, we use weight matrix to represent the graph

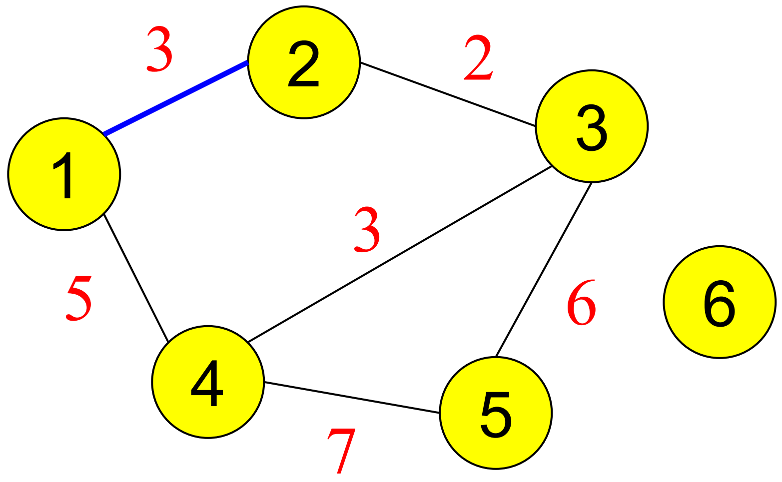
$$C = c[i, j], \quad i, j = 1, 2, \ldots, n,$$

where

$$c[i,j] = \begin{cases} c(i,j), & if\ (i,j) \in E \\ \theta, & if\ otherwises \end{cases}$$

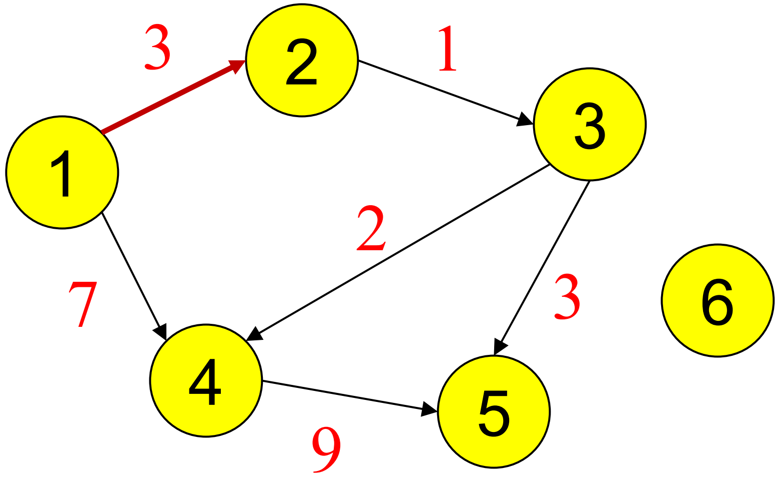
- θ: special value to identify (i, j) is not an edge; depends on the case, the value of θ could be: 0, $+\infty$, $-\infty$.

Weight matrix of undirected graph



$$\begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0 & 3 & 0 & 5 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 0 \\ 3 & 0 & 2 & 0 & 3 & 6 & 0 \\ 4 & 5 & 0 & 3 & 0 & 7 & 0 \\ 5 & 0 & 0 & 6 & 7 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

Weight matrix of directed graph



$$
\begin{array}{c|cccccc}
 & 1 & 2 & 3 & 4 & 5 & 6 \\
\hline
1 & 0 & 3 & 0 & 7 & 0 & 0 \\
2 & 0 & 0 & 1 & 0 & 0 & 0 \\
3 & 0 & 0 & 0 & 2 & 3 & 0 \\
4 & 0 & 0 & 0 & 0 & 9 & 0 \\
5 & 0 & 0 & 0 & 0 & 0 & 0 \\
6 & 0 & 0 & 0 & 0 & 0 & 0
\end{array}
$$

Graph Representation

1. Incidence matrix
2. Adjacency matrix
3. Weight matrix
4. **Adjacency list**

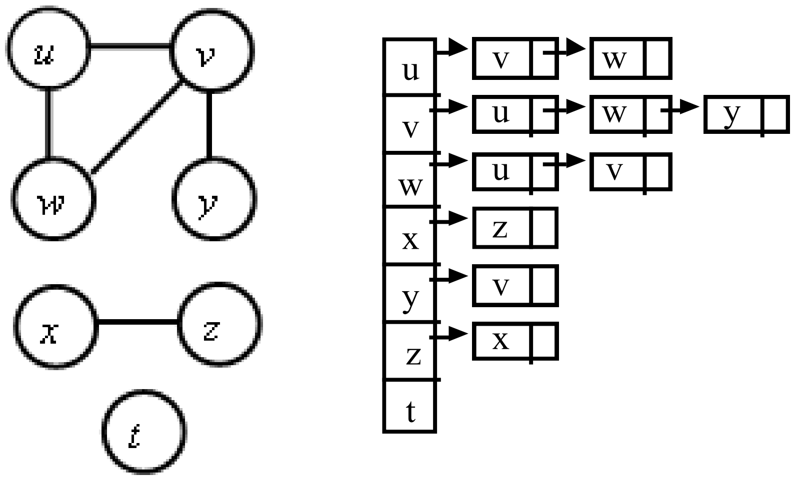
4. Adjacency List

Adjacency list: each vertex has a list of which vertices it is adjacent

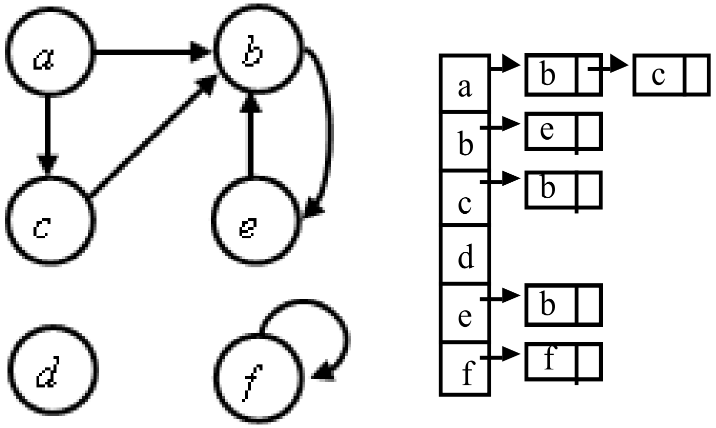
- Is an array Adjacency consiststing of $|V|$ list
- Each vertex has 1 list
- Each vertex $u \in V$: Adjacency[u] consists of nodes that are adjacent to u.

Example:

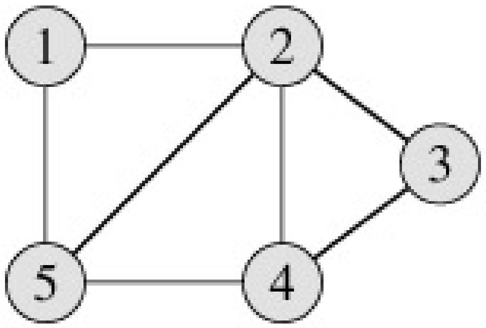
Undirected graph

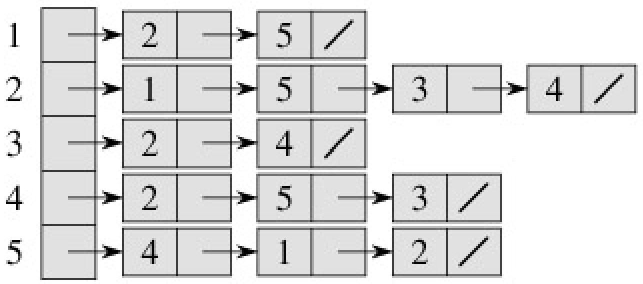


Directed graph



Graph representation



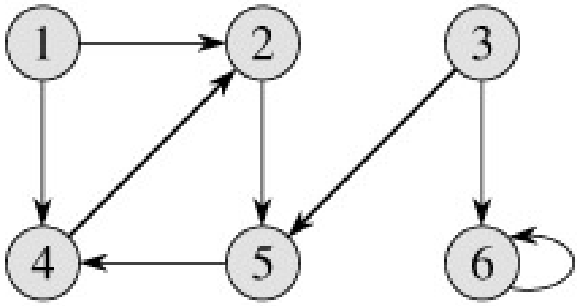


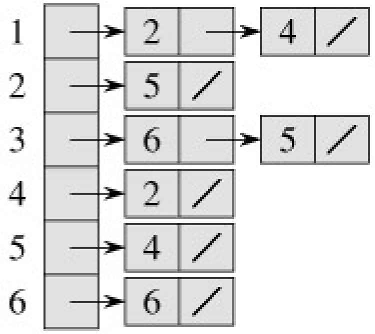
	1	2	3	4	5
1	0	1	0	0	1
2	1	0	1	1	1
3	0	1	0	1	0
4	0	1	1	0	1
5	1	1	0	1	0

graph

Adjacency list

Adjacency matrix





	1	2	3	4	5	6
1	0	1	0	1	0	0
2	0	0	0	0	1	0
3	0	0	0	0	1	1
4	0	1	0	0	0	0
5	0	0	0	1	0	0
6	0	0	0	0	0	1

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THANK YOU !